Class 13 Transcriptomics and the analysis of RNA-Seq data

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```
library(BiocManager)
```

```
library(DESeq2)
```

```
Loading required package: S4Vectors

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics


Attaching package: 'generics'

The following objects are masked from 'package:base':

    as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
    setequal, union


Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

    IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

    anyDuplicated, aperm, append, as.data.frame, basename, cbind,
    colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
    get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
    order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
    rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
    unsplit, which.max, which.min


Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

    findMatches
```

```
The following objects are masked from 'package:base':

    expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats


Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

    colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
    colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
    colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
    colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
    colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
    colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
    colWeightedMeans, colWeightedMedians, colWeightedSds,
    colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
    rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
    rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
    rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
    rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
    rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
    rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
    rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Welcome to Bioconductor

    Vignettes contain introductory material; view with
    'browseVignettes()'. To cite Bioconductor, see
    'citation("Biobase")', and for packages 'citation("pkgname")'.


Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

    rowMedians
```

```
The following objects are masked from 'package:matrixStats':

    anyMissing, rowMedians
```

Background

Today we will perform an RNASeq analysis of the effects of a common sertoid on airway cells.

In particular, dexamethasone (herafter just called “dex”) on different airway smooth muscle cell lines (ASM cells).

Data Import:

We need two different inputs:

- **countData**: with genes in rows and experiments in columns
- **colData**: meta data that desscribes the columns in countData

```
counts <- read.csv("airway_scaledcounts.csv", row.names= 1)
metadata <-  read.csv("airway_metadata.csv")
```

Wee peaks at counts and metadata

```
head(counts)
```

```
                SRR1039508 SRR1039509 SRR1039512 SRR1039513 SRR1039516
ENSG00000000003        723        486        904        445       1170
ENSG00000000005          0          0          0          0          0
ENSG00000000419        467        523        616        371        582
ENSG00000000457        347        258        364        237        318
ENSG00000000460         96         81         73         66        118
ENSG00000000938          0          0          1          0          2
                SRR1039517 SRR1039520 SRR1039521
ENSG00000000003       1097        806        604
ENSG00000000005          0          0          0
ENSG00000000419        781        417        509
ENSG00000000457        447        330        324
ENSG00000000460         94        102         74
ENSG00000000938          0          0          0
```

```
heads(metadata)
```

```
   group group_name      value
1      1         id SRR1039508
2      1         id SRR1039509
3      1         id SRR1039512
4      1         id SRR1039513
5      1         id SRR1039516
```

```
6       1          id SRR1039517
7       2         dex    control
8       2         dex    treated
9       2         dex    control
10      2         dex    treated
11      2         dex    control
12      2         dex    treated
13      3    celltype     N61311
14      3    celltype     N61311
15      3    celltype    N052611
16      3    celltype    N052611
17      3    celltype    N080611
18      3    celltype    N080611
19      4      geo_id GSM1275862
20      4      geo_id GSM1275863
21      4      geo_id GSM1275866
22      4      geo_id GSM1275867
23      4      geo_id GSM1275870
24      4      geo_id GSM1275871
```

> Q1. How many genes are in this dataset?

```
nrow(counts)
```

```
[1] 38694
```

> Q2. How many 'control' cell lines do we have?

```
sum(metadata$dex == "control")
```

```
[1] 4
```

Differential gene expression

We have 4 replicate drug treated and control (no drug) columns/experiments in our `counts` object.

We want one "mean" value for each gene (rows) in "treated" (drug) and one mean value for each gene in "control" cols.

Step 1. Find all "control" columms Step 2. Extract these columns to a new object called `control.counts` Step 3. Then calculate the mean value for each gene

Step 1.

```
control.inds <- metadata$dex == "control"
```

Step 2.

```
control.counts <- counts[ , control.inds]
```

Step 3.

```
control.mean <- rowMeans(control.counts)
```

> Q4. Follow the same procedure for the treated samples (i.e. calculate the mean per gene across drug treated samples and assign to a labeled vector called treated.mean)

Now do the same thing for the “treated” columns/experiments...

```
treated.inds <- metadata$dex == "treated"
```

Step 2.

```
treated.counts <- counts[ , treated.inds]
```

Step 3.

```
treated.mean <- rowMeans(treated.counts)
```

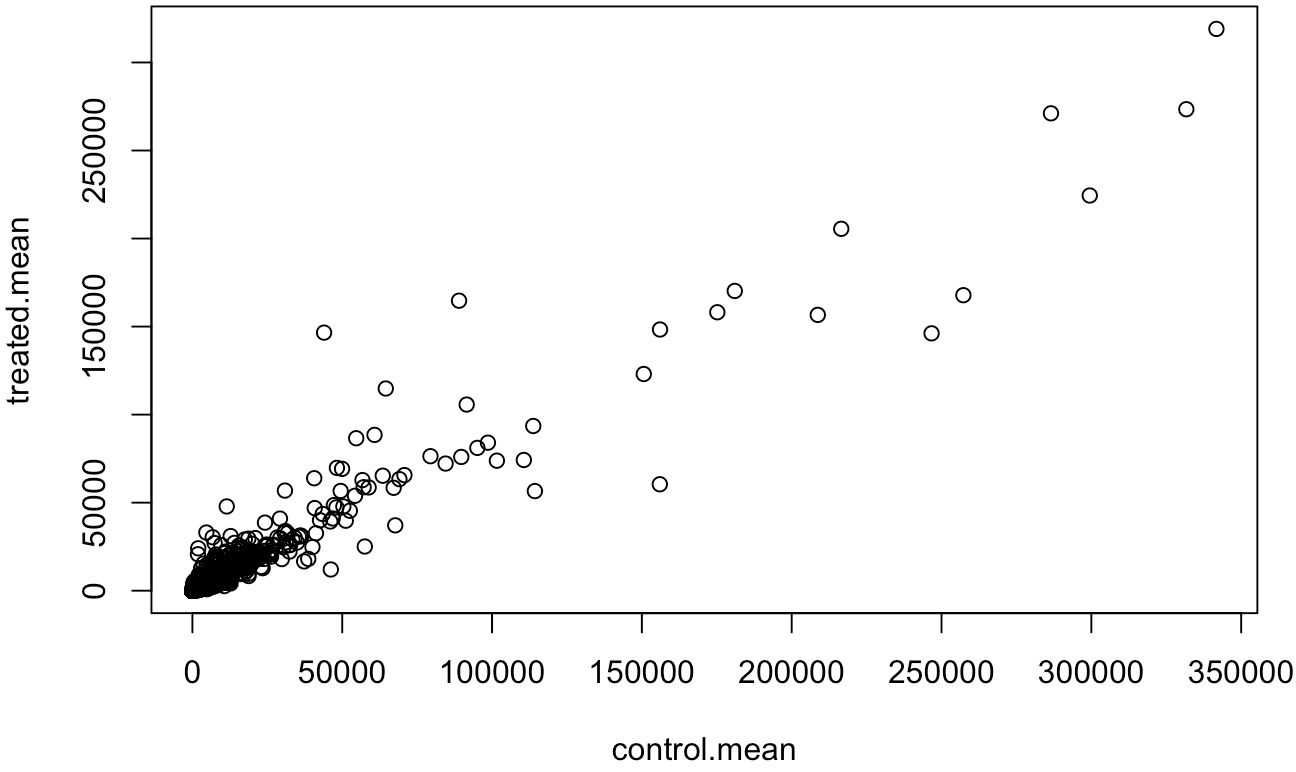
> Q. Now do the same thing for the “treated” columns/experiments...

Put these together for easy book-keeping as `meancounts`

```
meancounts <- data.frame(control.mean , treated.mean)
```

> Q5a. A quick plot

```
plot(meancounts)
```

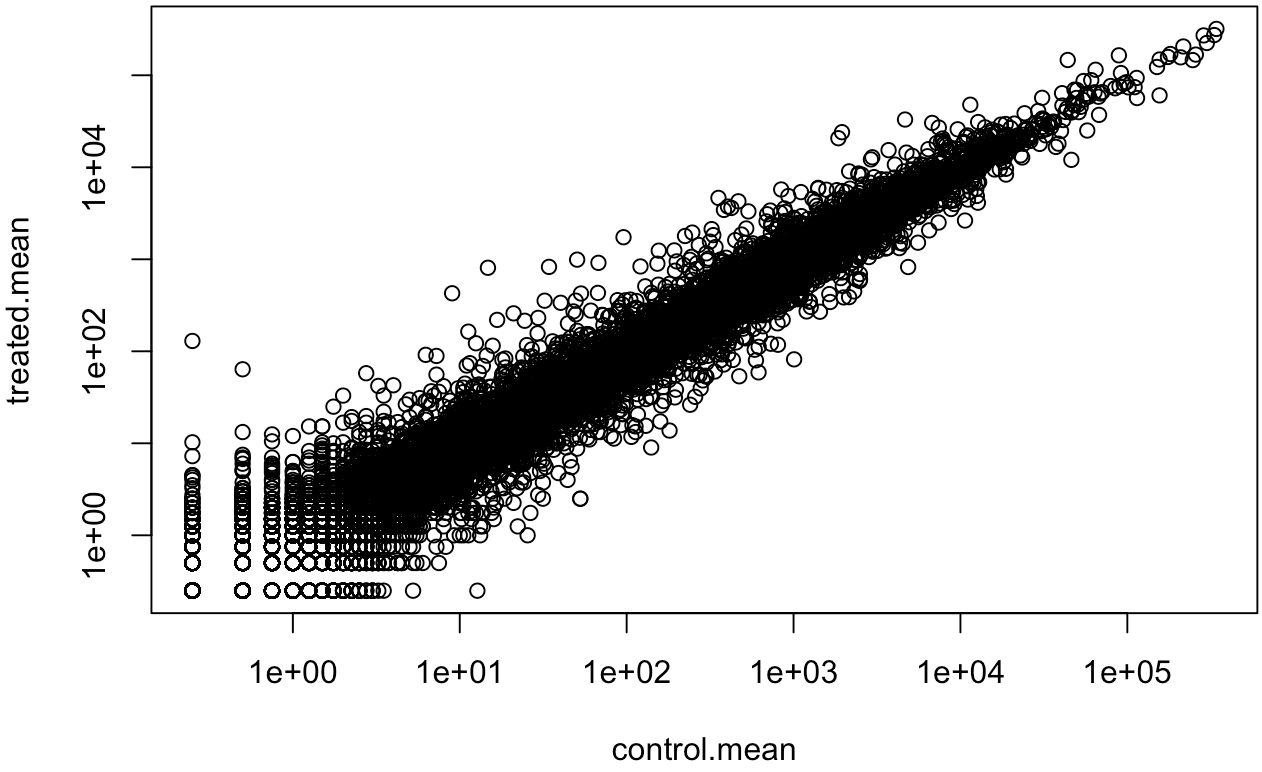


> Q6. Let's log transofrm this count data:

```
plot(meancounts, log="xy")
```

```
Warning in xy.coords(x, y, xlabel, ylabel, log): 15032 x values <= 0 omitted
from logarithmic plot

Warning in xy.coords(x, y, xlabel, ylabel, log): 15281 y values <= 0 omitted
from logarithmic plot
```



> Q5b. You could also use the ggplot2 package to make this figure producing the plot below. What geom_?() function would you use for this plot? geom_point()

N.B We most often use log2 for this type of data as it makes the interpretation much more straightforward.

Treated/Control is often called “fold-change”

If there was no change we would have a log2-fc of zero

```
log2(10/10)
```

```
[1] 0
```

If we had double the amount of transcript around

```
log2(20/10)
```

```
[1] 1
```

If we had half as much transcript around we would have a log2-fc of -1

```
log2(5/10)
```

```
[1] -1
```

> Q. Calculate a log2 fold change value for all our genes and add it as a new column to our `meancounts` object.

```
meancounts$log2f <- log2(meancounts$treated.mean/meancounts$control.mean)
head(meancounts)
```

```
                control.mean treated.mean       log2f
ENSG00000000003       900.75       658.00 -0.45303916
ENSG00000000005         0.00         0.00         NaN
ENSG00000000419       520.50       546.00  0.06900279
ENSG00000000457       339.75       316.50 -0.10226805
ENSG00000000460        97.25        78.75 -0.30441833
ENSG00000000938         0.75         0.00        -Inf
```

There are some “funky” log2fs values (NaN and -Inf) here that have come about whenever we have 0 mean count values. Typically we would remove these genes from any further analysis - as e can’t cay anything about them if we have no data for them.

> Q7. What is the purpose of the arr.ind argument in the which() function call above? Why would we then take the first column of the output and need to call the unique() function?

arr.ind = TRUE. We take the first column because we care about which rows are zeros. () to avoids counting the same gene multiple times.

```
head(meancounts[, 1:2] == 0)
```

```
                control.mean treated.mean
ENSG00000000003        FALSE        FALSE
ENSG00000000005         TRUE         TRUE
ENSG00000000419        FALSE        FALSE
ENSG00000000457        FALSE        FALSE
ENSG00000000460        FALSE        FALSE
ENSG00000000938        FALSE         TRUE
```

> Q8. Using the up.ind vector above can you determine how many up regulated genes we have at the greater than 2 fc level?

```
zero.vals <- which(meancounts[,1:2]==0, arr.ind=TRUE)

to.rm <- unique(zero.vals[,1])
```

```
mycounts <- meancounts[-to.rm,]
head(mycounts)
```

```
                control.mean treated.mean       log2f
ENSG00000000003       900.75       658.00 -0.45303916
ENSG00000000419       520.50       546.00  0.06900279
ENSG00000000457       339.75       316.50 -0.10226805
ENSG00000000460        97.25        78.75 -0.30441833
ENSG00000000971      5219.00      6687.50  0.35769358
ENSG00000001036      2327.00      1785.75 -0.38194109
```

```
up.ind <- mycounts$log2fc > 2
```

> Q9. Using the down.ind vector above can you determine how many down regulated genes we have at the greater than 2 fc level?

```
down.ind <- mycounts$log2fc < (-2)
```

> Q10. Do you trust these results? Why or why not?

No because we are not considering statistical significance.

##DESeq analysis

Let's do this analysis with an estimate of statistical significance using the **DESeq2** package. .

```
library(DESeq2)
```

DESeq (like many bioconductor packages) want it's input data in a very specific way.

```
dds <- DESeqDataSetFromMatrix(countData = counts,
                       col = metadata,
                      design= ~dex)
```

```
converting counts to integer mode

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
design formula are characters, converting to factors
```

Run the DESeq analysis pipeline

```
dds<-DESeq(dds)
```

```
estimating size factors

estimating dispersions
```

```
gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing
```

```
res <- results(dds)
head(res)
```

```
log2 fold change (MLE): dex treated vs control
Wald test p-value: dex treated vs control
DataFrame with 6 rows and 6 columns
                  baseMean log2FoldChange     lfcSE      stat    pvalue
                 <numeric>      <numeric> <numeric> <numeric> <numeric>
ENSG00000000003 747.194195      -0.350703  0.168242 -2.084514 0.0371134
ENSG00000000005   0.000000             NA        NA        NA        NA
ENSG00000000419 520.134160       0.206107  0.101042  2.039828 0.0413675
ENSG00000000457 322.664844       0.024527  0.145134  0.168996 0.8658000
ENSG00000000460  87.682625      -0.147143  0.256995 -0.572550 0.5669497
ENSG00000000938   0.319167      -1.732289  3.493601 -0.495846 0.6200029
                     padj
                <numeric>
ENSG00000000003  0.163017
ENSG00000000005        NA
ENSG00000000419  0.175937
ENSG00000000457  0.961682
ENSG00000000460  0.815805
ENSG00000000938        NA
```

```
36000 *0.05
```

```
[1] 1800
```

```
library(AnnotationDbi)
library(org.Hs.eg.db)
```

```
res$entrez <- mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     column="ENTREZID",
                     keytype="ENSEMBL",
                     multiVals="first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
res$uniprot <- mapIds(org.Hs.eg.db,
                      keys=row.names(res),
                      column="UNIPROT",
                      keytype="ENSEMBL",
                      multiVals="first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
res$genename <- mapIds(org.Hs.eg.db,
                      keys=row.names(res),
                      column="GENENAME",
                      keytype="ENSEMBL",
                      multiVals="first")
```

```
'select()' returned 1:many mapping between keys and columns
```

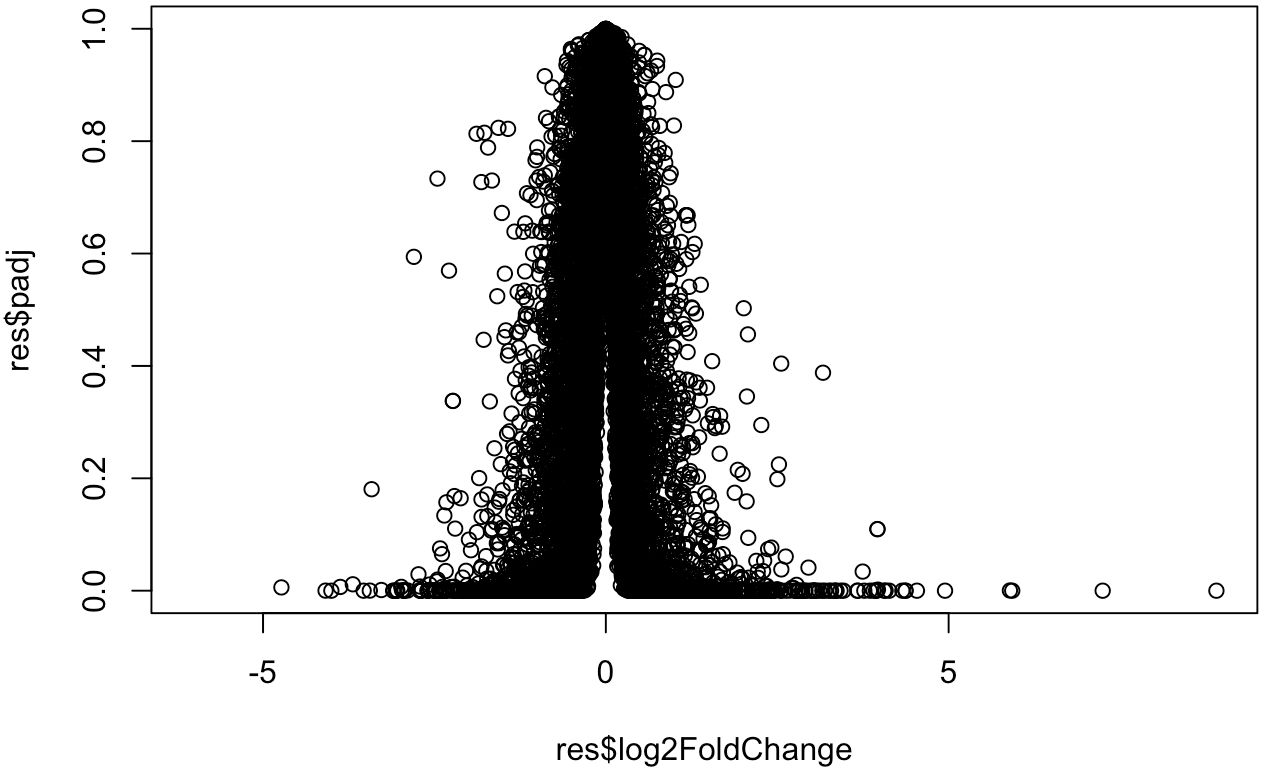
```
head(res)
```

```
log2 fold change (MLE): dex treated vs control
Wald test p-value: dex treated vs control
DataFrame with 6 rows and 9 columns
                  baseMean log2FoldChange     lfcSE      stat    pvalue
                 <numeric>      <numeric> <numeric> <numeric> <numeric>
ENSG00000000003 747.194195      -0.350703  0.168242 -2.084514 0.0371134
ENSG00000000005   0.000000             NA        NA        NA        NA
ENSG00000000419 520.134160       0.206107  0.101042  2.039828 0.0413675
ENSG00000000457 322.664844       0.024527  0.145134  0.168996 0.8658000
ENSG00000000460  87.682625      -0.147143  0.256995 -0.572550 0.5669497
ENSG00000000938   0.319167      -1.732289  3.493601 -0.495846 0.6200029
                     padj      entrez     uniprot               genename
                <numeric> <character> <character>            <character>
ENSG00000000003  0.163017        7105  A0A087WYV6          tetraspanin 6
ENSG00000000005        NA       64102      Q9H2S6            tenomodulin
ENSG00000000419  0.175937        8813      H0Y368 dolichyl-phosphate m..
ENSG00000000457  0.961682       57147      X6RHX1 SCY1 like pseudokina..
ENSG00000000460  0.815805       55732      A6NFP1 FIGNL1 interacting r..
ENSG00000000938        NA        2268      B7Z6W7 FGR proto-oncogene, ..
```

Volcano plot

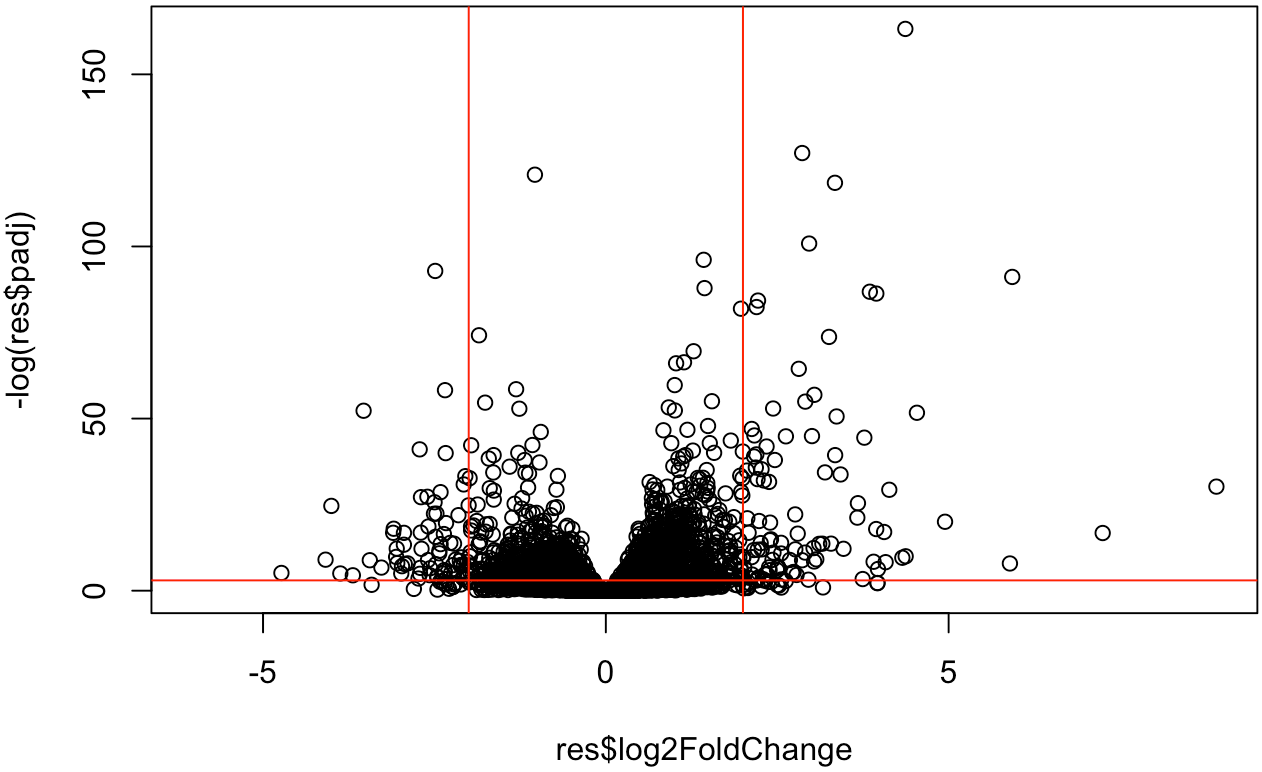
This is a main summary results figure form these kinds of studies. It is a plot of Log2-foldchange vs P-value.

```
plot(res$log2FoldChange,
     res$padj)
```



Again this y-axis is highly skewed and needs a log transformation plot and we can flip the y-axis with a minus sign so it looks like every other volcano plot.

```
plot(res$log2FoldChange,
     -log(res$padj))
abline(v=-2, col="red")
abline(v=+2, col="red")
abline(h=-log(0.05), col="red")
```



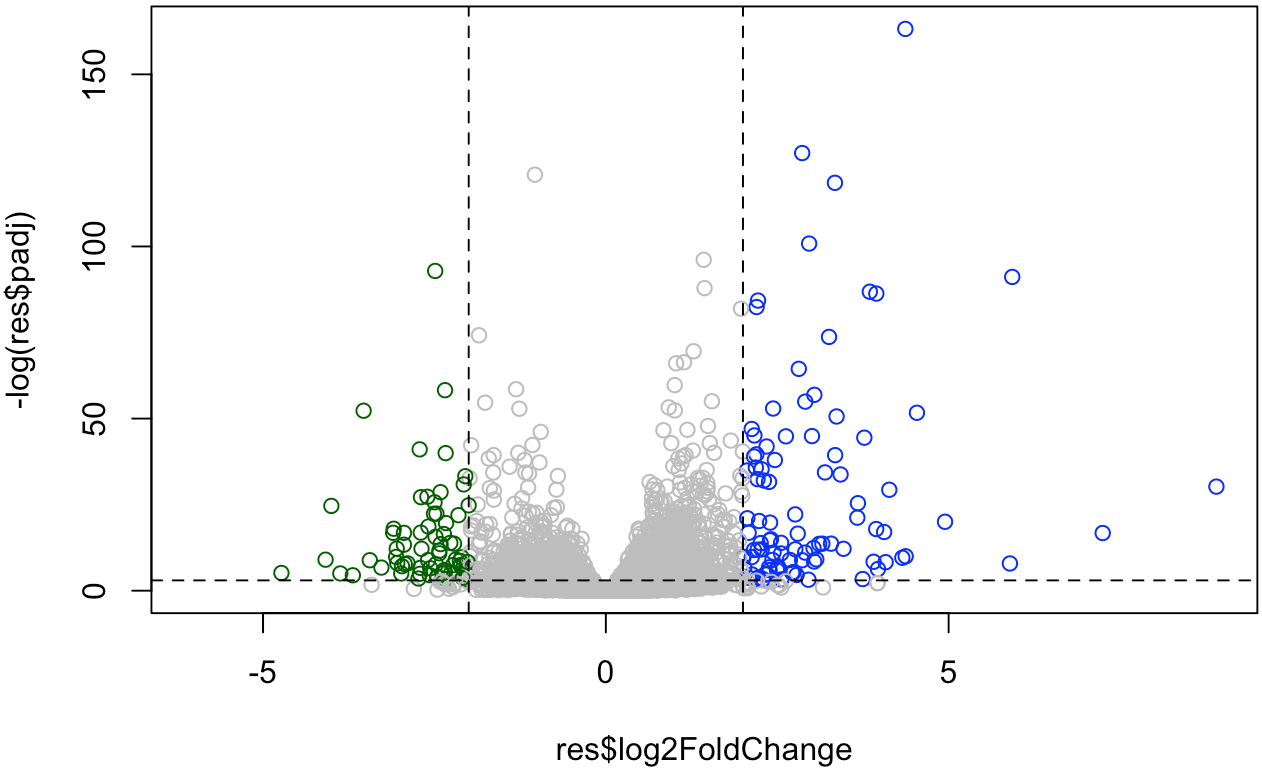
Adding some color annotation

Start with a default base color “gray”

```
#custom colors
mycols <- rep("gray", nrow(res))
mycols[res$log2FoldChange >2] <- "blue"
mycols[res$log2FoldChange < -2] <- "darkgreen"
mycols[res$padj >= 0.05 ] <- "gray"

#volcano plot
plot(res$log2FoldChange,
      -log(res$padj),
      col =mycols)

#cut-off dashed lines
abline(v=c(-2,+2), lty=2)
abline(h=-log(0.05), lty =2)
```

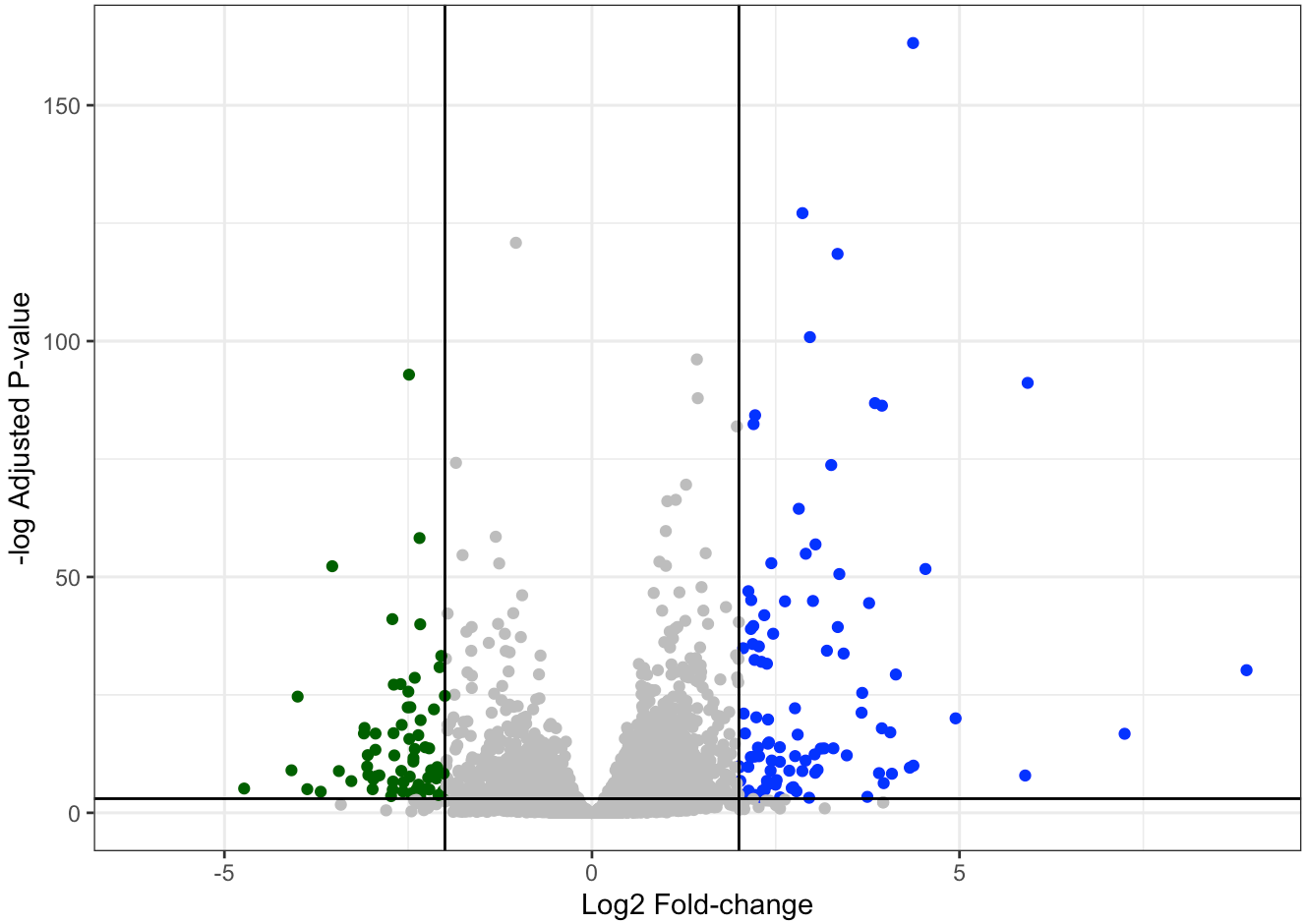


> Make a presentation quality ggplot version of this plot. Include clear axis labels, a clean theme, your custom colors, cut-off lines and a plot title.

```
library(ggplot2)

ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
  geom_point(colour = mycols) +
  labs(x="Log2 Fold-change",
       y="-log Adjusted P-value") +
geom_vline(xintercept = c(-2,2)) +
  geom_hline(yintercept = -log(0.05)) +
theme_bw()
```

```
Warning: Removed 23549 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Save our results

Write a CSV file

```
write.csv(res, file="results.csv")
```

Add annotation data

We need to add missing annotation data to our main `res` results object. This includes the common gene "symbol"

```
head(res)
```

```
log2 fold change (MLE): dex treated vs control
Wald test p-value: dex treated vs control
DataFrame with 6 rows and 9 columns
                  baseMean log2FoldChange     lfcSE      stat    pvalue
                 <numeric>      <numeric> <numeric> <numeric> <numeric>
ENSG00000000003 747.194195      -0.350703  0.168242 -2.084514 0.0371134
ENSG00000000005   0.000000             NA        NA        NA        NA
```

```
ENSG00000000419 520.134160       0.206107  0.101042  2.039828 0.0413675
ENSG00000000457 322.664844       0.024527  0.145134  0.168996 0.8658000
ENSG00000000460  87.682625      -0.147143  0.256995 -0.572550 0.5669497
ENSG00000000938   0.319167      -1.732289  3.493601 -0.495846 0.6200029
                     padj      entrez     uniprot               genename
                <numeric> <character> <character>            <character>
ENSG00000000003  0.163017        7105  A0A087WYV6          tetraspanin 6
ENSG00000000005        NA       64102      Q9H2S6            tenomodulin
ENSG00000000419  0.175937        8813      H0Y368 dolichyl-phosphate m..
ENSG00000000457  0.961682       57147      X6RHX1 SCY1 like pseudokina..
ENSG00000000460  0.815805       55732      A6NFP1 FIGNL1 interacting r..
ENSG00000000938        NA        2268      B7Z6W7 FGR proto-oncogene, ..
```

We will use R and bioconductor to do this "ID mapping"

```
library("AnnotationDbi")
library("org.Hs.eg.db")
```

Let's see what databases we can use for translation/mapping...

```
columns(org.Hs.eg.db)
```

```
 [1] "ACCNUM"       "ALIAS"        "ENSEMBL"      "ENSEMBLPROT"  "ENSEMBLTRANS"
 [6] "ENTREZID"     "ENZYME"       "EVIDENCE"     "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"     "GO"           "GOALL"        "IPI"          "MAP"
[16] "OMIM"         "ONTOLOGY"     "ONTOLOGYALL"  "PATH"         "PFAM"
[21] "PMID"         "PROSITE"      "REFSEQ"       "SYMBOL"       "UCSCKG"
[26] "UNIPROT"
```

We can use the `mapIds()` function now to translate between any of these databases.

```
res$symbol <- mapIds(org.Hs.eg.db,
                     keys=row.names(res), # Our genenames
                     keytype="ENSEMBL",        # The format of our genenames
                      column="SYMBOL")          # The new format we want to add
```

```
'select()' returned 1:many mapping between keys and columns
```

> Q. Also add "ENTREZID", "GENENAME"

```
res$entrez <- mapIds(org.Hs.eg.db,
                     keys=row.names(res), # Our genenames
                     keytype="ENSEMBL",        # The format of our genenames
                      column= "ENTREZID")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
res$genename <- mapIds(org.Hs.eg.db,
                       keys=row.names(res), # Our genenames
                       keytype="ENSEMBL",        # The format of our genenames
                        column= "GENENAME")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
head(res)
```

```
log2 fold change (MLE): dex treated vs control
Wald test p-value: dex treated vs control
DataFrame with 6 rows and 10 columns
                  baseMean log2FoldChange     lfcSE      stat    pvalue
                 <numeric>      <numeric> <numeric> <numeric> <numeric>
ENSG00000000003 747.194195      -0.350703  0.168242 -2.084514 0.0371134
ENSG00000000005   0.000000             NA        NA        NA        NA
ENSG00000000419 520.134160       0.206107  0.101042  2.039828 0.0413675
ENSG00000000457 322.664844       0.024527  0.145134  0.168996 0.8658000
ENSG00000000460  87.682625      -0.147143  0.256995 -0.572550 0.5669497
ENSG00000000938   0.319167      -1.732289  3.493601 -0.495846 0.6200029
                     padj      entrez     uniprot               genename
                <numeric> <character> <character>            <character>
ENSG00000000003  0.163017        7105  A0A087WYV6          tetraspanin 6
ENSG00000000005        NA       64102      Q9H2S6            tenomodulin
ENSG00000000419  0.175937        8813      H0Y368 dolichyl-phosphate m..
ENSG00000000457  0.961682       57147      X6RHX1 SCY1 like pseudokina..
ENSG00000000460  0.815805       55732      A6NFP1 FIGNL1 interacting r..
ENSG00000000938        NA        2268      B7Z6W7 FGR proto-oncogene, ..
                     symbol
                <character>
ENSG00000000003      TSPAN6
ENSG00000000005        TNMD
ENSG00000000419        DPM1
ENSG00000000457       SCYL3
ENSG00000000460       FIRRM
ENSG00000000938         FGR
```

> Q11. Run the mapIds() function two more times to add the Entrez ID and UniProt accession and GENENAME as new columns called res$entrez$, res$uniprot and res$genena

```
res$entrez <- mapIds(org.Hs.eg.db,
                     keys = row.names(res),
                     keytype = "ENSEMBL",
                     column = "ENTREZID",
                     multiVals = "first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
res$uniprot <- mapIds(org.Hs.eg.db,
                      keys = row.names(res),
                      keytype = "ENSEMBL",
                      column = "UNIPROT",
                      multiVals = "first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
res$genename <- mapIds(org.Hs.eg.db,
                       keys = row.names(res),
                       keytype = "ENSEMBL",
                       column = "GENENAME",
                       multiVals = "first")
```

```
'select()' returned 1:many mapping between keys and columns
```

##Save annotated results to a CSV file

```
write.csv(res, file="results_annotated.csv")
```

##Pathway analysis

What known biological pathways do our differentialy expressed genes overlap with (i.e. play a role in)?

There are lots of bioconductor packages to do this type of analysis.

We will use one of the oldest called **gage** along with **pathview** to render nice pics of the pathways we find.

We can install these with the command: BiocManager::install(c("pathview", "gage", "gageData"))

```
library(pathview)
library(gage)
library(gageData)
```

Have a wee peak what is in `gageData`

```
# Examine the first 2 pathways in this kegg set for humans
data(kegg.sets.hs)
head(kegg.sets.hs, 2)
```

```
$`hsa00232 Caffeine metabolism`
[1] "10"   "1544" "1548" "1549" "1553" "7498" "9"

$`hsa00983 Drug metabolism - other enzymes`
 [1] "10"     "1066"   "10720"  "10941"  "151531" "1548"   "1549"   "1551"
 [9] "1553"   "1576"   "1577"   "1806"   "1807"   "1890"   "221223" "2990"
[17] "3251"   "3614"   "3615"   "3704"   "51733"  "54490"  "54575"  "54576"
[25] "54577"  "54578"  "54579"  "54600"  "54657"  "54658"  "54659"  "54963"
```

```
[33] "574537" "64816"  "7083"   "7084"   "7172"   "7363"   "7364"   "7365"
[41] "7366"   "7367"   "7371"   "7372"   "7378"   "7498"   "79799"  "83549"
[49] "8824"   "8833"   "9"      "978"
```

The main `gage()` function that does the work wants a simple vector as input.

```
foldchanges <- res$log2FoldChange
names(foldchanges) <- res$symbol
head(foldchanges)
```

```
     TSPAN6        TNMD        DPM1       SCYL3       FIRRM         FGR
-0.35070296          NA  0.20610728  0.02452701 -0.14714263 -1.73228897
```

The KEGG database uses ENTREZ ids so we need to provide these in our input vector for **gage**:

```
names(foldchanges) <- res$entrez
```

Now we can use **gage**

```
# Get the results
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

What is in the output object `keggres`

```
attributes(keggres)
```

```
$names
[1] "greater" "less"    "stats"
```

```
# Look at the first three down (less) pathways
head(keggres$less, 3)
```

```
                                     p.geomean stat.mean        p.val
hsa05332 Graft-versus-host disease 0.0004250607 -3.473335 0.0004250607
hsa04940 Type I diabetes mellitus  0.0017820379 -3.002350 0.0017820379
hsa05310 Asthma                    0.0020046180 -3.009045 0.0020046180
                                       q.val set.size         exp1
hsa05332 Graft-versus-host disease 0.09053792       40 0.0004250607
hsa04940 Type I diabetes mellitus  0.14232788       42 0.0017820379
hsa05310 Asthma                    0.14232788       29 0.0020046180
```

We can use the **pathview** function to render a figure of any of these pathways along with annotation for our DEGs

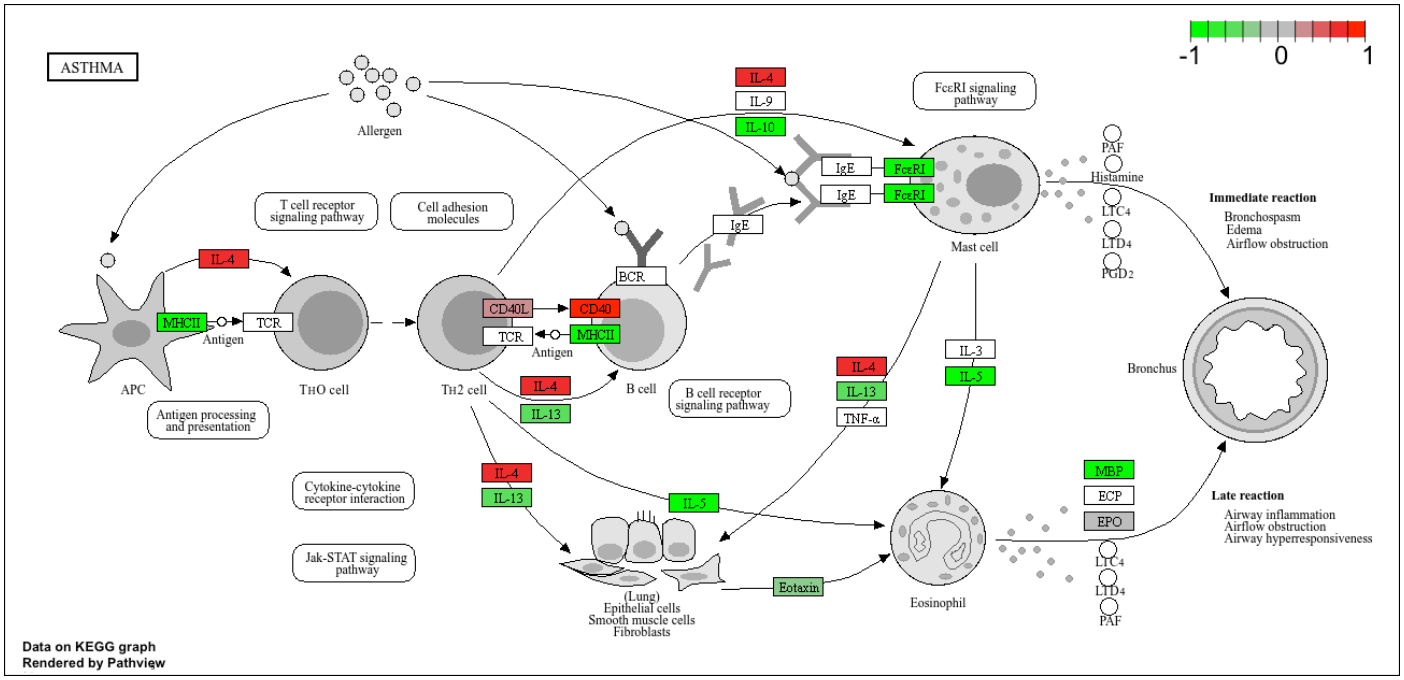
Let's see the hsa05310 Asthma pathway with our DEGs colored up:

```
pathview(gene.data=foldchanges, pathway.id="hsa05310")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/erin/Documents/BIMM 143/class13/class13

Info: Writing image file hsa05310.pathview.png
```



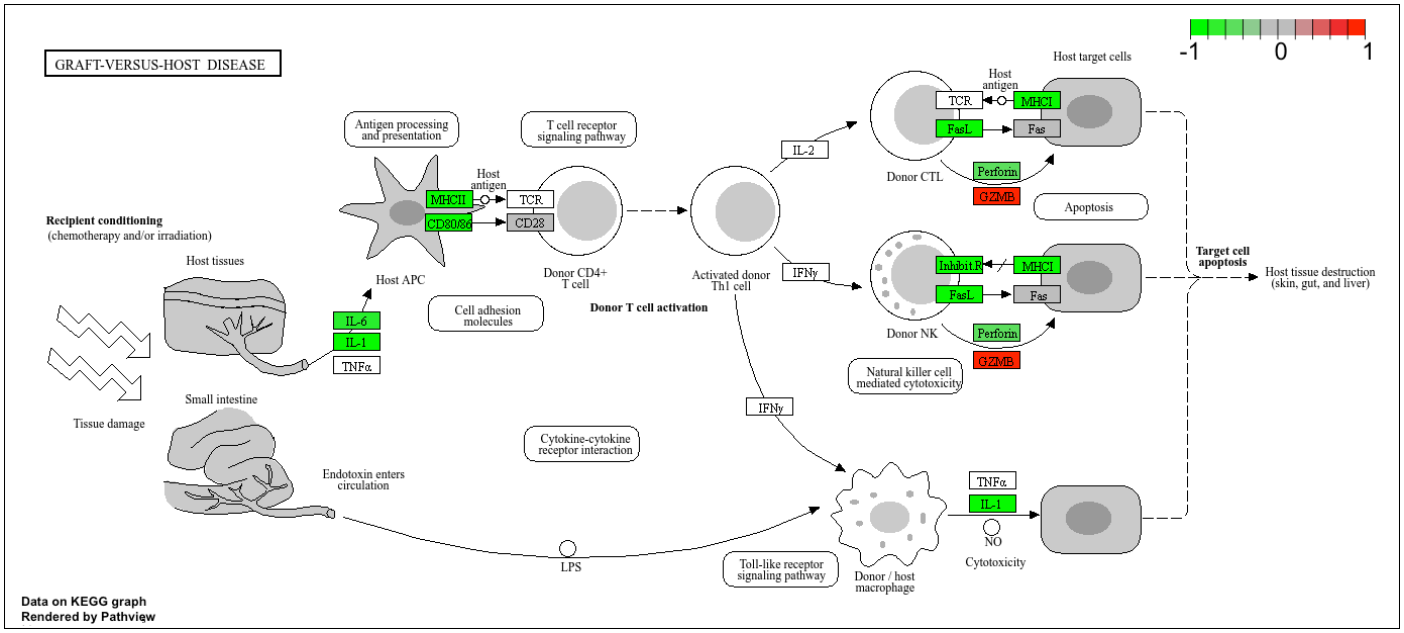
> Q12. Can you do the same procedure as above to plot the pathview figures for the top 2 down-reguled pathways Graft-versus-host disease" and "Type I diabetes"?

```
pathview(gene.data=foldchanges, pathway.id="hsa05332")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/erin/Documents/BIMM 143/class13/class13

Info: Writing image file hsa05332.pathview.png
```

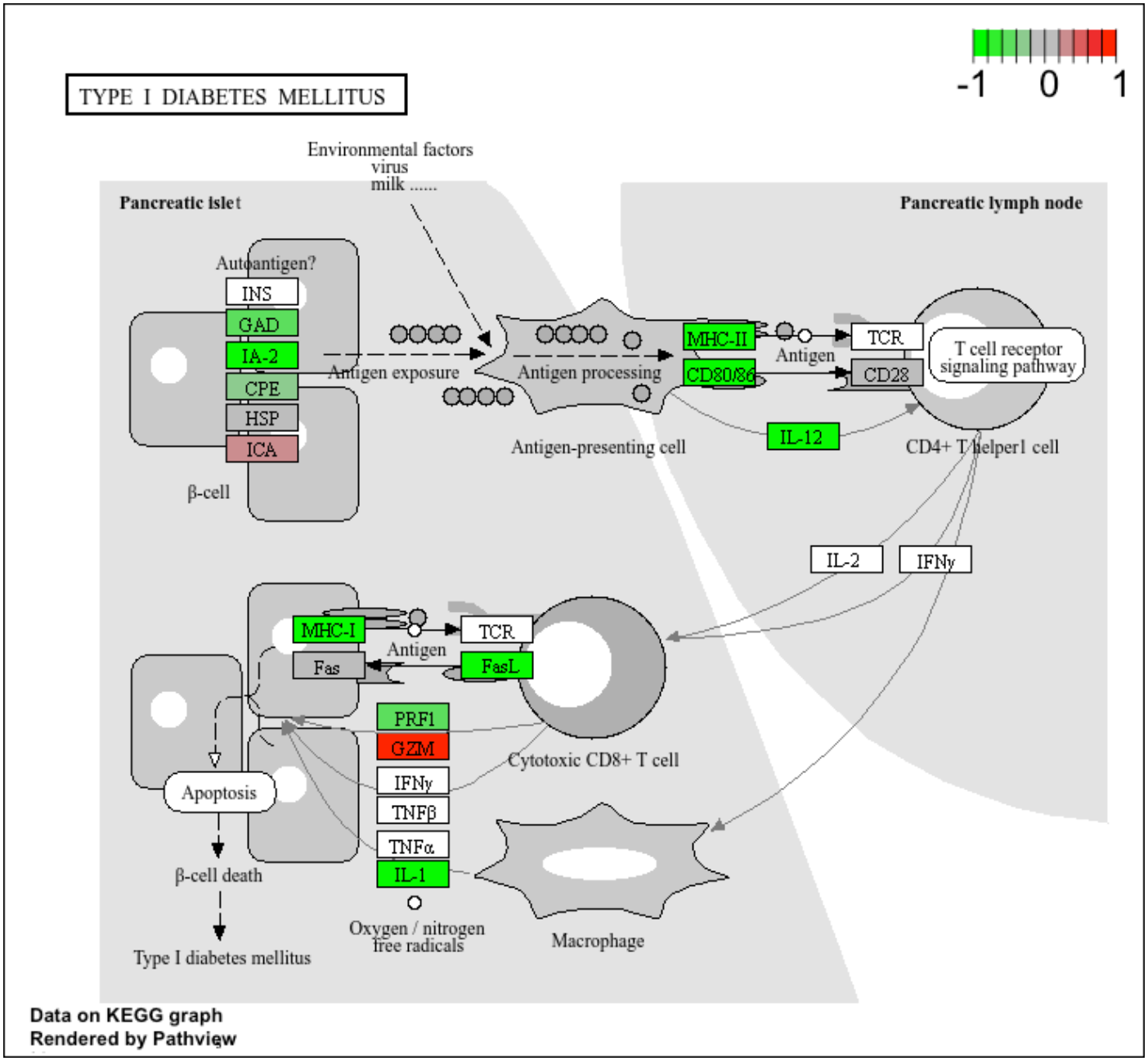



```
pathview(gene.data=foldchanges, pathway.id="hsa04940")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/erin/Documents/BIMM 143/class13/class13

Info: Writing image file hsa04940.pathview.png
```


-1 0 1
TYPE I DIABETES MELLITUS
Environmental factors
virus
milk .....
Pancreatic islet
Pancreatic lymph node
Autoantigen?
INS
GAD
IA-2
CPE
HSP
ICA
β-cell
Antigen exposure
Antigen processing
MHC-II
CD80/86
Antigen
TCR
CD28
T cell receptor signaling pathway
IL-12
Antigen-presenting cell
CD4+ T helper1 cell
IL-2
IFNγ
MHC-I
Antigen
TCR
Fas
FasL
PRF1
GZM
IFNγ
TNFβ
TNFα
IL-1
Cytotoxic CD8+ T cell
Apoptosis
β-cell death
Type I diabetes mellitus
Oxygen / nitrogen
free radicals
Macrophage
Data on KEGG graph
Rendered by Pathview